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DOSE, TIME AND FRACTIONATION: A CLINICAL HYPOTHESIS

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Based on published clinical results of radiotherapy, a formula is suggested which relates total dose, number of fractions and overall treatment time to a quantity termed 'Nominal Standard Dose'. This quantity represents the biological effect of a given treatment regime. Using this concept it is possible to compare various treatment schedules that involve different fractionation patterns and various overall treatment times. The evidence upon which the idea is based, and also its use in routine clinical practice, are discussed.

In clinical radiotherapy difficulties amounting to confusion arise because of the unsuitability for comparison of one technique with another. The usual statement, made because of simplicity, is either in terms merely of total dose (6000 rad) or in terms of dose and time (6000 rad in 5 weeks). From radiobiological work (Ellis, 1967) and from clinical experience, there is no doubt that the dose per fraction and interval between fractions as well as the related parameters of number of fractions, total dose and total time are all of importance to the final biological effect. If one figure could be used representing a course of radiation which reaches normal tissue tolerance, it would clearly be advantageous in comparing techniques.

This figure should represent the *normal connective tissue tolerance* because this is the limiting factor in most tumour therapy. Treatments which do not reach tolerance should be expressed as a proportion of this figure. For instance (see later) if the normal tissue tolerance with 30 fractions of 200 rad given daily in 40 days is represented by a

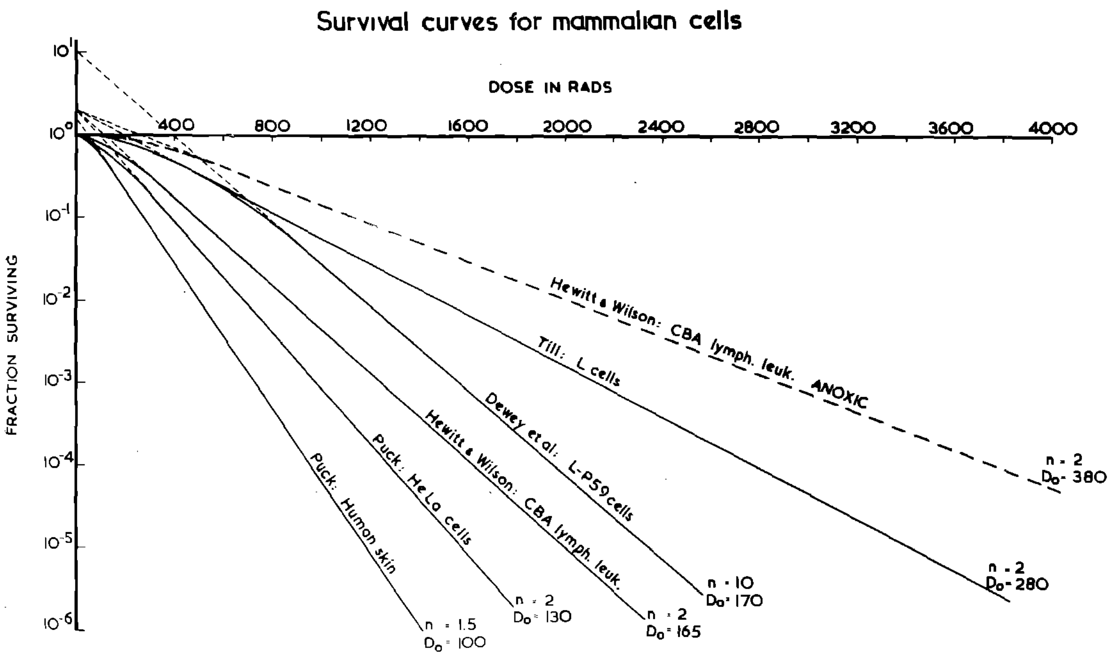

FIG. 1

Collection of published survival curves for mammalian cells cultured *in vitro* and *in vivo* to illustrate the wide range of survival curve parameters observed.

figure of 1765, then a course using only part of this tolerance (say 18 fractions) would be represented by a figure of $\frac{18}{30} \times 1765 = 1060$.

Some normal tissues are more sensitive than others.

Possibly the most sensitive normal non-genetic tissue is the kidney (this may be partly because of its firm capsule tending to exacerbate the effect of any swelling of tissue inside it). In giving a single session of 200 rad to a tumour in, say, the para-aortic region, suppose the dose to the kidney is 85 rad. It might be that to reach normal connective tissue tolerance with such small individual daily fractions of 85 rad about 110 fractions would be needed. If this is the case, then the 30 fractions would represent $\frac{30}{110} = 0{\cdot}26$ of the connective tissue tolerance $= 580$ and not, as might be otherwise supposed, $\frac{85}{200}$ $(=0{\cdot}43 \times 1740)$. In cases where the type of technique used might depend on kidney 'dose effect' such considerations are clearly of importance. It follows that if the normal connective tissue tolerance is the limiting factor in treatment, then the effect on the tumour of a course of treatment is incidental. In view of the difficulty – even impossibility – of knowing, in any individual case, what is the rate of division of the cells of the tumour under conditions of treatment, it follows that the treatment should be given to the tumour as quickly as possible, compatible with the connective tissue tolerance and 'patient tolerance'. This latter depends on both local and systemic reactions.

LIMITATIONS OF LABORATORY EVIDENCE

Radiotherapeutic dosage must be precise. Errors of 20% in dosage can be clinically detected and it has been shown (Morrison & Deeley, 1962) that the effects of a change in dosage to patients of 10% produce statistically significant differences in recurrence rate. Radiobiological evidence has been of inestimable value in demonstrating the uniform *qualitative* effects of radiation but their *quantitative* application to radiotherapy is not justified.

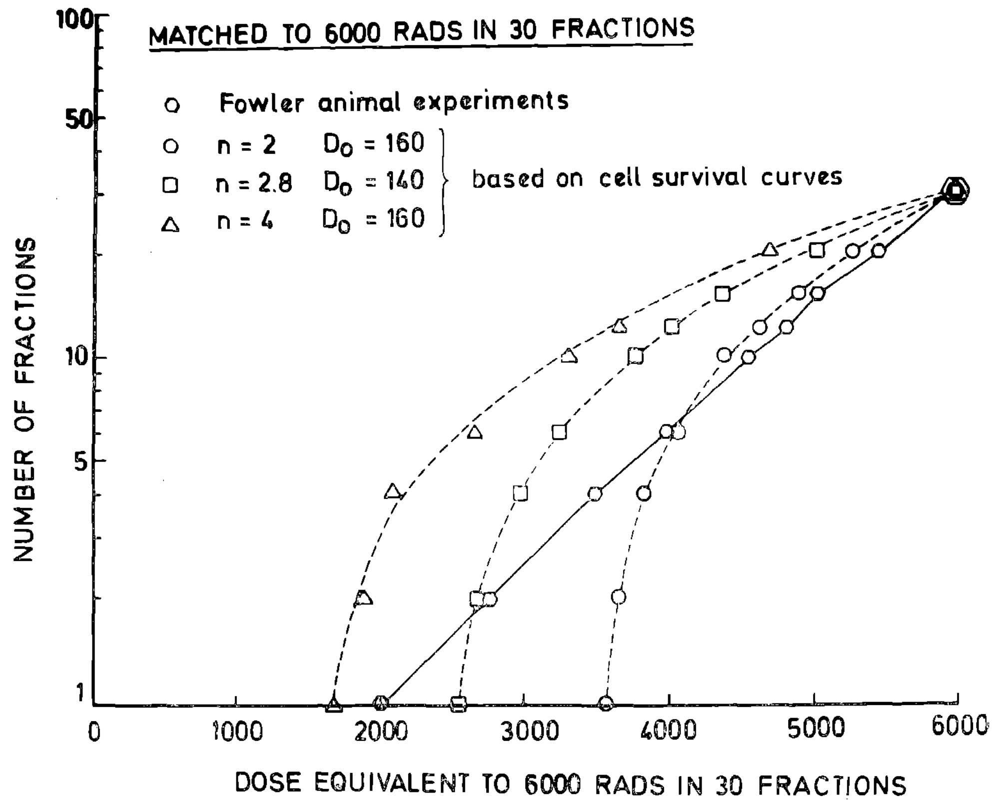

FIG. 2

The hexagons refer to Fowler's animals experiments and are iso-effect curves relating total dose necessary to produce a given effect to the number of fractions. The other three curves are theoretical iso-effect curves calculated from three possible survival curves showing that small changes in survival curve parameters have a large effect on the iso-effect curve.

(From R. Oliver, Personal Communication)

The most important concepts introduced into the thinking of radiotherapists as a result of radiobiological research are as follows:

(a) Cell survival curves indicate that, in general, a certain dose of radiation always kills the same proportion of the cell population.

(b) Small doses of radiation are less efficient in terms of cell killing because of the 'shoulder' on the cell survival curve.

(c) Under conditions of 'hypoxia' cells are less sensitive to radiation in the ratio of about 3 to 1, than when they are efficiently oxygenated.

(d) Vascular changes occur in a tumour during a course of fractionated treatment resulting in the conversion of hypoxic cells to a state of oxygenation.

(e) The effect of a given physical absorbed dose is not the same for all types of radiation.

The characteristics of different cell-survival curves as published in the literature, however, are different (Fig. 1). The importance of these differences is indicated in Fig. 2 where it is shown that relatively small differences in the survival curve parameters can result in large differences in quantitative effects. Cells can also differ in other respects; for example, Fig. 3 shows how two

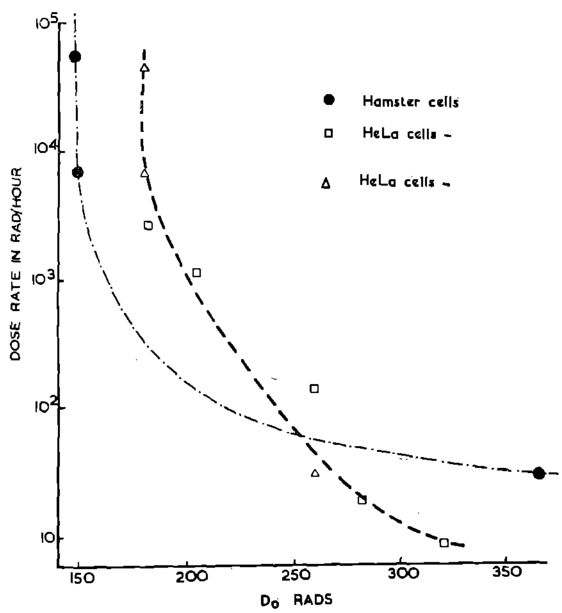

FIG. 3

Illustrating the change of the 37% dose-slope (D_0) of the survival curve with changing dose-rate for two mammalian cell lines cultured *in vitro*. The dose-rate effect is quantitatively different for the two cell lines. (Unpublished data due to Dr. E. J. Hall).

different cell lines respond differently to changes of dose-rate. Because of these differences the use of any one set of cell survival data as a basis for clinical radiotherapy does not seem to be justified. Pig skin experiments carried out by Fowler and his colleagues have provided further evidence of an animal clinical-radiobiological type. This evidence confirms the clinical evidence that, for the same visible effect, if the number of fractions is smaller the total dose must be smaller. Fowler (1965) gives data based on pig skin experiments to be applied quantitatively to clinical skin reactions in radiotherapy.

EVIDENCE FROM CLINICAL MATERIAL

Because of the difficulties for clinical purposes, in using radiobiological data to relate dose, time and fractions to effect, the author decided to use human data, based on the earlier reasoning and assumptions in this paper.

(a) **Skin tolerance.** In 1942 Ellis published data based on human skin reactions. Associated with these was the hypothesis that because:

(i) the healing of skin epithelium depends on the condition of the underlying connective tissue stroma (whether dermis or subcutaneous tissue),

(ii) apart from bone and brain, connective tissues throughout the body are similar,

(iii) within and around a malignant tumour normal connective tissue elements make up the stroma.

Therefore, apart from bone and brain, the 'tumour dose' limited by the normal tissue tolerance dose, could be based on skin tolerance.

In Fig. 4 (Cohen 1960) the two lower graphs represent Strandqvist-type relations between log dose and log time for the skin tolerance and skin erythema. In addition to the author's (1942) data the skin tolerance line is based on data by Paterson (1948) and Jolles (1946). The lowest line is based on clinical experiments by Reisner (1932) and by Quimby (1936).

It seems that both have the same slope (0·33) suggesting that damage and repair in skin for widely differing end points, depend on the same mechanisms – *cell killing, intracellular recovery and homeostatically (extracellular) controlled repair.*

(b) **Squamous carcinoma cure**

In the same figure (4) the upper graph shows the relationship between log-dose and log-time for published 5 year survivals by many authors and of various types (Ellis 1967, quoting from Cohen 1960). Its slope is 0·22.

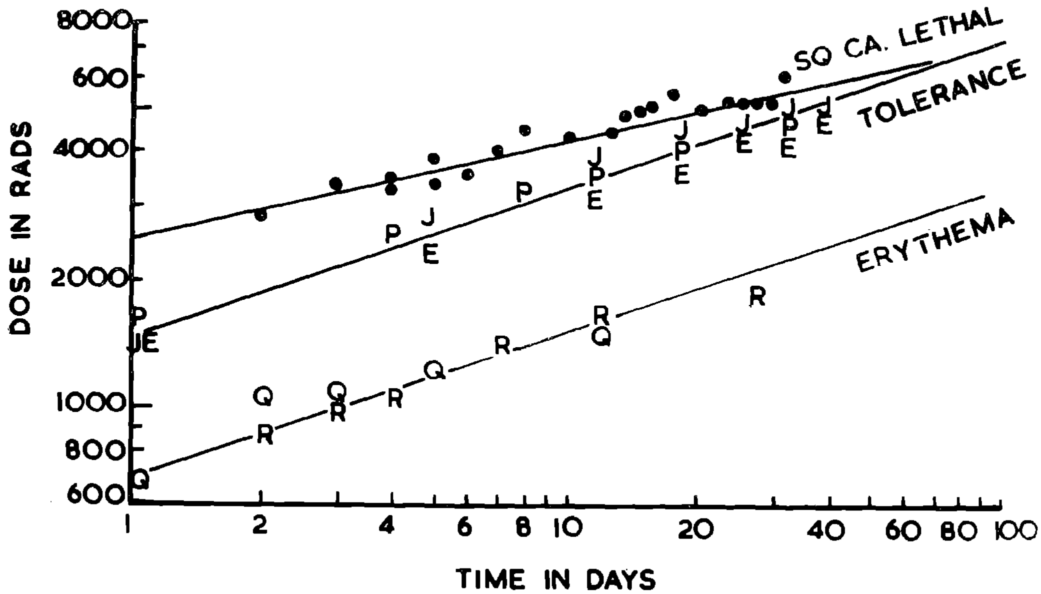

FIG. 4

Iso-effect curves for different fractionation regimes for squamous cell carcinoma, for skin erythema and for normal tissue tolerance. Taken from Cohen (1960).

It is justifiable to assume that:

(1) Squamous carcinoma is not subject to homeostatic control.

(2) That squamous carcinoma is a cell type, therefore, in which the effect depends on the number of fractions and their magnitude, and on internal recovery, i.e. the effect depends only on the radiation damage and internal recovery of the cells.

DEVELOPMENT OF THE HYPOTHESIS

(a) Assumptions

The hypothesis is based on the following assumptions:

(i) Malignant cells are not susceptible to homeostatic control as are normal cells. Such control is complex in nature and may involve effects, among others, due to immune factors, histiocyte migration and hormone control. We know that some tumours are hormone sensitive to some extent. In-so-far as they are sensitive to hormones they are not behaving like malignant cells.

(ii) Homeostatic factors depend on time. The stimulus for homeostatic control passes to various parts of the body which respond at varying rates to the death and autolysis of normal cells by repair process to the tissues.

(iii) All cells which are damaged by radiation, whether benign or malignant, either become incapable of reproduction or undergo a recovery process from sublethal damage.

(b) Separation of fractionation and time factors

From the above it is seen that the difference between the tumour effect and the normal skin effect is the participation of homeostatic control in the skin effects. As this has been related to time (v. supra) it is considered that the difference between the two slopes (0·11) in Fig. 4 represents a regression coefficient with time for normal tissue recovery while the regression coefficient 0·22 represents the effect of fractionation. If it is assumed (justifiably) that the techniques used employed 5 fractions per week then by plotting log-dose against log-number-of-fractions, the graph has a regression coefficient of 0·24 (R. Oliver, Personal Communication).

From this the author suggests that the following formula represents the effect on normal tissues:

$$\text{Total dose} = \text{N.S.D. } N^{0\cdot 24}.\ T^{0\cdot 11} \text{—(1)}$$

– where the Total Dose is in rad, N = number of fractions, T = elapsed time in days and N.S.D. = Nominal Standard Dose – in ret (rad equivalent therapy – analagous to, but more precise than, the rem.)

The *N.S.D.* requires some explanation – it is not considered that it represents a single dose since such a dose (lasting say 5 minutes) could not have a time measured satisfactorily in days. If the time were to be 1 day (=24 hrs.) 2 doses would be required. Therefore it is considered that it is a parameter representing quantitatively a biological effect.

It is considered necessary that *all values of N.S.D. should be related to normal tissue tolerance* conditions since there is not a linear relationship between dose and effect.

If 18 fractions of 300 rad are given on a schedule of 3 times weekly, the total dose is 5400 rad and the NSD may be calculated from equation (1) to be 1800 ret. If equation (1) is applied to the first 9 fractions given in 18 days the NSD appears to be 1160 ret.

$$(\text{since } \frac{2700}{9^{0\cdot24} \cdot 18^{0\cdot11}} = 1160)$$

Two halves of such a course summated would thus amount to $2 \times 1160 = 2320$ ret. This is incorrect since the dose in equation (1) is a full tolerance dose, which 2700 rad clearly is not. A figure representing the half treatment course cited in the example should be 9/18 of the N.S.D. i.e. 900 ret. The suggested term for this figure is *partial tolerance*.

CLINICAL USE OF THE HYPOTHESIS

(a) **Irregularities in Treatment**

By the use of this concept the biological effect of a course of treatment in the following practical situations can be represented by a single figure in each case–

(a) When there is an interruption in the treatment schedule e.g. split course.
(b) When there is a change in the dose per fraction.
(c) The incidental effect on a critical organ.

(b) **Planning of a Treatment Schedule** – can be based on the N.S.D. having regard to the local tissue tolerance, to patient tolerance and to the end effect on connective tissue.

(i) Local tissue tolerance; if the mucosa in, say, the pharynx becomes ulcerated and a patient cannot swallow during a course of treatment, the treatment schedule should be adjusted so that the local conditions are tolerable, i.e. so that the balance between destruction and repair of epithelium is maintained without reducing the final biological effect on the connective tissue.

(ii) Patient tolerance, nausea, vomiting, malaise, blood count etc. can make it necessary to modify a schedule in an individual case or as a planned series. The same final, local, biological effect on the normal connective tissue in the tumour area can still be the aim but smaller fractions over a long time and with different intervals may be necessary.

(iii) Tumour tolerance – if we could know the cell cycle parameters of the normal and malignant cells during a course of treatment in every case it might be possible to plan and carry out a course of treatment accordingly. But as we know none of these the *general aim* must be to *achieve the normal tissue tolerance effect in as short a time as possible.* The reason for keeping the time short is to crowd as much radiation into as few cycles of the tumour cells as possible. In cases where there is reason from clinical evidence to suppose that the tumour cells divide infrequently, the treatment schedule may be prolonged to suit the social and economic situation (e.g. it was possible in the palliative treatment of a recurrent carcinoma of the rectum to treat a man as an outpatient twice weekly for 13 weeks with a good local palliative result. He was able to work and maintain his family throughout the course).

SUPPORT FOR THE HYPOTHESIS

Support for this hypothesis comes from its clinical use and from suggestions by Fowler (1965):

(a) **Clinical evidence** (i) In clinical practice it is possible to achieve similar clinical effects with very different treatment schedules e.g. 6 daily fractions of 515 rad with X-rays in one week, and 6 weekly fractions of 615 rad with X-rays in 5 weeks.

(ii) Clinical schedules for patients treated to tolerance levels in various centres give comparable and rather similar values for the NSD. See Table I.

(iii) Assuming that under treatment conditions, only internal recovery (dependent on the number and magnitude of fractions) is of importance to the tumour cells, the relationship for these becomes Total Dose = Tumour Effect. $N^{0\cdot24}$. It is found, for instance, that iso-effect tumour curves by Scott (1961) and by Friedman, Pearlman and Turgeon (1967) for Hodgkins disease give tumour effect figures corresponding as in Table 2.

(b) **Pig-skin evidence**

The work of Fowler and his colleagues (1965) on pig-skin combined with clinical evidence has prompted the suggestion of the figures given in Column I of Table 3. The fractions were all given over six weeks. It can be seen that there is virtual coincidence of the N.S.D. values down to 4 fractions.

CONCLUSION

The N.S.D. hypothesis is a suitable concept for clinical work, providing a basis for comparison within and between centres of the effect of different

TABLE 1

TABLE OF NSD VALUES IN CLINICAL CASES

Location of tumour	Total dose (rad)	No. of Fractions/ Days	Target Volume (cm^3)	NSD (normal tissue) (ret)	NSD (tumour) (ret)
Centre V					
Larynx	7200	26/35	200	2220	3290
Larynx	6500	,,	,,	2010	2970
Centre W					
Brain	6000	15/21	220	2240	3140
Bladder	6000	,,	700	2240	3140
Larynx	5750	,,	125	2150	3019
Larynx (revised)	5500	,,	,,	2060	2878
Bladder (revised)	5500	,,	700	2060	2878
Brain (revised)	4500	,,	220	1680	2353
Centre X					
Larynx (cured)	6000	15/20	350	2250	3140
Larynx (cured)	5400	16/21	700	1990	2770
Centre Z					
Pharynx S*	6000	20/42	600	1940	2930
Larynx S*	5000	13/28	125	1870	2700
Ca. rectum (inop.)	6000	18/75	6000	1870	3000
Bladder S*	5200	13/49	500	1830	2800
Thyroid S*	4950	13/33	3000	1820	2670
Wilm's tumour	6300	24/98	3600	1780	2950
Breast	4500	9/70	2400	1660	2660
Piriform sinus and nodes S*	4875	16/40	650	1670	2510
Breast	3200	4/21	2400	1640	2285
Breast	3600	6/35	2400	1590	2340
Breast	3800	10/21	2400	1570	2180
Post-cricoid F*	4200	21/28	400	1400	2020
Cerebellar astrocytoma	3900	13/84	800	1300	2100

* S = Survival for 5 years. F = Treatment failed

These NSD values are calculated from the formula

$NSD = \text{Total Dose} \times N^{-0.24} \times T^{-0.11}$

Incomplete treatments in Centre Z are not calculated as proportions of the intended full treatment, as would be necessary if additional doses were to be considered

TABLE 2

TUMOUR NSD'S IN HODGKIN'S DISEASE

	Friedman et al, 1967		Scott 1961	
No. of fractions, N	Total Dose, D (rad)	Tumour NSD $= \frac{D}{N^{0.24}}$ (ret)	Total Dose, D (rad)	Tumour NSD $= \frac{D}{N^{0.24}}$ (ret)
1	1200	1200		
4	1870	1340	1600	1150
8	2130	1300		
12	2430	1330	2200	1200
15	2600	1350		
19	2720	1340		
23	2830	1330	2900	1360
26	2920	1330		
29	3000	1320		

Doses given are Minimum corroborated tumour lethal doses

TABLE 3

VALUES OF N.S.D. BASED ON DOSES SUGGESTED BY FOWLER (DOSES GIVEN IN SIX WEEKS).

Number of Fractions	Total doses from Animal Experiments and Modal Clinical Data (rad)	N.S.D. (ret)
30	6000	1775
20	5460	1790
15	5070	1760
12	4810	1775
10	4560	1750
8	4300	1750
6	4010	1740
4	3490	1675
2	2730	1550
1	2000	1340

schedules on the basis of a single figure. It might be thought that instead of the N.S.D. as a figure for normal tissue tolerance we might use 100% or Unity. This would be possible if dosimetry and judgement were the same from one centre to another; but, as they are not, to use the N.S.D. is better. In association with computer analysis of recorded data from many centres it might even be possible in the future to detect differences between centres in dosimetry and in clinical assessment of tolerance.

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NOTICES

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ONE week Visitors' Courses are held at the Christie Hospital twice a year, in November and May. These courses are primarily intended for visitors from this country and overseas who already have some knowledge of radiotherapy but who wish to see current methods and techniques being used in Manchester. They may be useful for D.M.R.T. and F.F.R. examination candidates but are *not* meant to be revision courses for either examination. During these courses there will be a full programme of talks, discussions and demonstrations by all the consultant radiotherapy staff of the hospital, which will include all the common varieties of disease treated by radiotherapy. The formal course will last five days beginning on the first Mondays in November and May. Visitors may, if they wish, remain for one further week to attend normal Out-patient, x-ray therapy, Radium Theatre and Mould Room work.

No fee is asked for these courses but numbers will be limited. Prospective visitors are asked to write to Dr. E. C. Easson, Director of Radiotherapy, Christie Hospital, Withington, Manchester, 20, preferably at least one month in advance for further particulars.

THE USE OF COMPUTERS IN RADIOLOGY

SEPTEMBER 1969

The Royal Belgian Society of Radiology, under the auspices of the European Association of Radiology, is organising a Colloquium on this subject to be held in Brussels during September 1969. Members of the Institute have been invited to participate and anyone wishing to present a paper should contact the British Liaison Officer, Dr. Margaret Snelling at the Middlesex Hospital, London W.1.